

Koa Shen

Robotics Technician Candidate · Assembly · Repair · Electromechanical Testing

(415) 810-8344 koashensf@gmail.com linkedin.com/in/koa-shen/ github.com/koa-shen koashen.com

EDUCATION

University of California, Santa Barbara

Expected June 2028

B.S. Mechanical Engineering · 3.97 major / 3.94 overall GPA · Dean's Honors every quarter

Santa Barbara, CA

- Coursework: Dynamics, Mechanics of Materials, Circuits. Current: Dynamical Systems and Machine Learning.

EXPERIENCE

Maintenance Technician

Summer 2025

Exploratorium

San Francisco, CA

- Serviced 100+ interactive exhibits; diagnosed failures and fabricated replacement parts using machine-shop, welding, and woodshop tools.
- Led three exhibit overhauls, including a mechanism redesign that avoided \$3,000+ in replacement costs; documented CAD for future repair.

Fabrication Team Lead

Fall 2024 – Present

Gaucha Racing (FSAE EV) — UC Santa Barbara

Santa Barbara, CA

- Led 15 teammates through design, fabrication, and installation of driver-safety hardware; passed FSAE technical inspection.
- Machined 33 steel and aluminum suspension components, holding critical bearing interfaces within 0.0005 in.

Mechanical Engineering Intern

Summer 2026 – Present (extended to part-time remote)

SLAC National Accelerator Laboratory — LCLS

Menlo Park, CA

- Built a Python/Drake digital twin and reusable stage catalog for high-DOF motion assemblies; collision checks run in milliseconds.
- Produced Solid Edge designs with GD&T and Teamcenter documentation; checked stage and mount loads with statics and dynamics calculations.

SELECTED PROJECTS

Compact Cycloidal Reducer & Stepper Package

Summer 2026 – Present

- Assembled and bench-tested a NEMA 17 stepper, TMC2209 driver, 15:1 reducer, and AS5600 encoder; validated feedback and missed-step detection.
- 3D-printed reducer components in PETG/ABS and built C++ firmware for step control and automatic position-error correction.

FSAE Steering Wheel Development

Fall 2024 – Spring 2026

- Integrated buttons, potentiometers, display, and quick-release into an FSAE steering wheel; iterative, FEA-guided redesign reduced mass from 5 lb to 1.7 lb.

TECHNICAL SKILLS

Maintenance & Fabrication: Mechanical repair, Failure diagnosis, Manual/CNC machining, Welding, 3D printing (PETG/ABS)

Robotics & Electronics: Mechanical assembly, Stepper drivers, Encoder/driver wiring, Arduino, C/C++, PlatformIO

CAD & Documentation: Solid Edge, SolidWorks, GD&T, Teamcenter PDM, Reverse engineering